INCH-POUND

MIL-H-27269A(USAF)
AMENDMENT 3
27 January 1989
SUPERSEDING
AMENDMENT 2
28 August 1981

MILITARY SPECIFICATION

HORIZONTAL SITUATION INDICATOR
GROUP AF/24J-1

This amendment forms a part of MIL-H-27269A(USAF), dated 20 December 1977, and is approved for use within the Department of the Air Force and is available for use by all Departments and Agencies of the Department of Defense.

PAGE 1

* 2.1 Delete title and first sentence and add: "Government documents."

* Add: "2.1.1 Specifications and Standards. The following specifications and standards, form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2)."

* Under SPECIFICATIONS, FEDERAL, delete "QQ-P-416 and title."

Delete address on lines 3 and 4 of beneficial comments block and add: "Oklahoma City Air Logistics Center/MMEDO, Tinker AFB OK 73145-5990."

* Change Federal Supply Class (FSC) from 6610 to 6605.

PAGE 2

* Under SPECIFICATIONS, MILITARY, delete "MIL-E-5272 and title."

* Under SPECIFICATIONS, MILITARY, MIL-L-25467, Change title to read:"Lighting, Integral Red, Aircraft Instrument, General Specification For" and add "MIL-C-83488 Coating, Aluminum, Ion Vapor Deposited."

* Under STANDARDS, MILITARY, delete "MIL-STD-143","MIL-STD-794" and add:"MIL-STD-781 Reliability Design Qualification And Production Acceptance Tests Exponential Distribution," "MIL-STD-810 Environmental Test Methods" and "MIL-STD-970 Standards And Specifications, Order of Preference For The Selection Of."

* Delete the parenthetical source statement that follows the listing of Government specification and standards and add: "(Unless otherwise indicated copies of federal and military specifications and standards are available from the Naval Publications and Forms Center, (ATTN: NPODS), 5801 Tabor Avenue, Philadelphia Pa 19120-5099.)"

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* Add "2.2 Non-Government. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issue of the documents cited in the solicitation (6.2)."

American Society For And Materials (ASTM)

ASTM D3951 Packing, Commercial

(Application for copies should be addressed to: ASTM, 1916 Race St, Philadelphia PA 19103.)

(Non-government standards and other publications are normally available from the organizations that prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)"*Add: "(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Naval Publications and Forms Center, (ATTN: NPODS), 5801 Tabor Avenue, Philadelphia PA 19120-5099.)"

* Add: "2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein (except for related associated detail specifications, specification sheets or MS standards), the text of this specification take precedence. Nothing in this specification, however, supersedes applicable laws and regulations unless a specific exemption has been obtained."

PAGE 3

3.1 Delete in its entirety and add: "3.1 First article. When specified (see 6.3) a sample shall be subjected to first article inspections (see 6.3) in accordance with 4.5."

* 3.3 Line 3 delete "MIL-STD-143" and add: "MIL-STD-970."

3.6,e Delete "Spray" and add: "Fog".

3.6,f Delete "Sand and Dust" and add: "Dust (fine sand)" in two places.

PAGE 6

Add: "3.6,o Mean-time-between-failures (MTBF). The horizontal situation indicator group shall have a MTBF of not less than 775 hours (O_o) at a confidence factor of 90 percent and discrimination ratio of 2. Testing to verify this requirement shall be in accordance with MIL-STD-781, Test Plan III C (see 4.4.24.2)."

PAGE 11

* 3.10 Line 5 delete "MIL-E-5272" and add: "MIL-STD-810".

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Figure 5: Delete Figure 5 and add: New Figure 5 attached in this amendment.

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* 3.15.1 Line 3 delete "position in" and add: "positioning".

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* 3.16.6.2 Delete "command" and add: "common".

* 3.16.6.3 Line 8 delete the sentence "Remove voltage A to C of RW-1 and apply it to B to C of card."

* 3.16.6.3 Line 11 delete "V to C" and add: "B to C".

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* 3.29.1 Line 2 delete "MIL-A-8626" and add: "MIL-A-8625".

Delete 3.29.2 in its entirety and add: "3.29.2 Steel parts. Steel parts shall be coated with ion vapor deposited aluminum, where practicable, in accordance with MIL-C-83488, type I or II as applicable and of a class that is adequate to achieve the degree of protection required. Other protective coating, in lieu of MIL-C-83488, may be used if demonstrated to be satisfactory and approved by the cognizant engineering activity. Cadmium plating must be avoided when satisfactory alternative processes can be used."

PAGE 28

Add: "3.33 Recycled and reclaimed materials. Recycled and reclaimed material shall be encouraged to the maximum extent possible without jeopardizing the intended end use of the item".

PAGE 29

* Delete "4.1", "a" and "b" in its entirety and add: "4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to performany of the inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements."

* Add: "4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspections, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material."

Delete "4.2 "a" and "b" in its entirety and add: "4.2 Classification of

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inspections. The inspection requirements specified herein are classified as follows:

a. Quality conformance inspection (see 4.2.1).

b. First article inspection (see 4.5)."

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* 4.2.2.2,h Delete "Salt spray" and add: "Salt fog".

* 4.2.2.2,i Delete "Sand and dust" and add: "Dust (fine sand)".

* 4.2.2.2.1 Under Group II delete "Salt spray" and add: "Salt fog".

* 4.2.2.2.1 Under Group II delete "Sand and dust" and add: "Dust (fine sand)".

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* Delete "4.3" in its entirety and add: "4.3 Inspection conditions. Unless otherwise specified all inspections shall be performed in accordance with the test condition specified in 4.3.1 of this specification.

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* 4.4.3.3 Line 3 delete "and recorded".

* 4.4.3.3 Line 7 delete recorded."

* 4.4.3.3 Line 9 delete "At end of paragraph", add:."

* 4.4.3.3 Line 10 move to end of paragraph. "In any sampling plan A or B test which requires repeating the scale error test, the azimuth card tolerance shall be ± 1.0 at each heading."

"PAGE 36

4.4.15 Delete existing paragraph and add: "4.4.15 Vibration error. The indicator shall be mounted, in its normal operating positions, on a vibration fixture, and properly connected for operations. While being operated, the indicator shall be vibrated with circular motion in a plane inclined 45 degrees to the horizontal with a total vibration amplitutude between 0.003 and 0.005 inch. The frequency of applied vibration shall be varied slowly from 5 to 50 cycles per second. While being vibrated, the indicator shall meet the following requirements."

PAGE 38

4.4.22 Line 3 Delete "MIL-E-5272" and add: "MIL-STD-810".

4.4.22.1 Line 1 add: "Method 501.1" before "Procedure I".

4.4.22.2 Delete entire paragraph and add: "4.4.22.2 Temperature altitude. The horizontal situation indicator group shall be subjected to a

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temperature-altitude test in accordance with Method 504.1 of MIL-STD-810 for equipmentcategory 6. There shall be no binding or sticking at any possible indication. After the horizontal situation indicator group has returned to room temperature, it shall meet all individual tests. There shall be a 5-minute warm-up period after the application of power before data is taken."

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4.4.22.3 Add "Method 503.1" before "Procedure I".

4.4.22.4 Add "Method 507.1" before "Procedure I".

4.4.22.5 Add "Method 508.1" before "Procedure I".

4.4.22.6 Title delete "Salt Spray" and add: "Salt fog." Add "Method 509.1" before "Procedure I" in line 1.

4.4.22.7 Title delete "Sand and Dust" and add: "Dust (fine sand)". Add "Method 510.1" before "Procedure I" in line 1.

4.4.22.8 Delete entire paragraph and add: "4.4.22.8 Vibration failure. The shall be subjected to a vibration type failure test as follows: The indicator shall be mounted on the vibration apparatus in its normal operating position. It shall be properly interconnected for operation. While being operated, it shall be vibrated with a circular motion of 0.018 to 0.020 inch diameter in a plane inclined 45 degrees to the horizontal plane and the frequency of vibration shall be varied uniformly form 5 to 50 cycles per second and return once each hour for a 3 hour period. Following the vibration test, the horizontal situation indicator group shall meet the requirements as specified in 4.4.4 through 4.4.8. No loosenese in the mechanism nor damage to any part of the horizontal situation indicator group shall result from the vibration."

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Add: "4.4.24.2 Mean-time-between-failure (MTBF). Equipment on-time accumulated during sampling/life tests shall be used in computation of MTBF. Accept/reject criteria shall be as specified in MIL-STD-781, Test Plan III C."

* 4.4.25 Line 8 delete " in accordance with MIL-D-4815".

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* 4.5 Delete " Qualification Testing" and add: "First article inspection".

* 4.5.1 Line 1 delete "Qualification Test Samples" and add: "First article samples".

* 4.5.2 Delete entirely and add: "4.5.2 First article required. Prior to actual procurement, the product which this specification covers shall pass the First Article tests specified herein. If the product is later modified in any way, the modified from shall be subjected to and shall pass the same First Article tests."

* 4.5.3 Delete entirely and add: "4.5.3 First article inspection. The first article inspections shall consist of all the tests specified herein."

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* Delete "4.6", "4.6.1" and "4.6.2."

* Add: "4.6 Inspection of packaging. The inspection of the preservation and interior package marking shall be in accordance with group A and B quality conformance inspection requirements, section 4 of MIL-P-116. The sampling and inspection of the packing and marking for shipment and storage shall be in accordance with the quality assurance provisions of applicable container specification and the marking requirements of MIL-STD-129."

* Delete Section 5 in its entirety and add:

"5. Packaging

5.1 Preservation. Preservation shall be level A, C, or industrial as specified. (see 6.2).

5.1.1 Level A.

5.1.1.1 Cleaning. Horizontal situation indicator shall be cleaned in accordance with process C-1 of MIL-P-116.

5.1.1.2 Drying. Horizontal situation indicator shall be dried in accordance with process D-4 of MIL-P-116.

5.1.1.3 Preservation application. Not applicable.

5.1.1.4 Unit Packaging. Unless otherwise specified by the contracting activity, each horizontal situation indicator shall be packaged in quantity unit packs of one each in accordance with Method IC-1 of MIL-P-116. Overbox completed pack in PPP-B-636 container. Apply sufficient cushioning material between bag and unit container of a type, density, and thickness to insure shock transmission does not exceed peak values in G's established for the horizontal situation indicator when completed packs are subjected to the rough handling drop tests of MIL-P-116.

5.1.2. Level C. Each horizontal situation indicator shall be clean, dry, and individually packaged in a manner that will afford adequate protection against corrosion, deterioration, and physical damage during shipment from supply source to the first receiving activity.

5.1.3. Industrial. The Industrial preservation of horizontal situation indicator shall be in accordance with ASTM D3951.

5.2. Packing. Packing shall be level A, B, C, or Industrial as specified (see 6.2).

5.2.1. Level A. Horizontal situation indicator packaged as specified in 5.1.1 shall be packed in shipping containers conforming to PPP-B-601, Styles A or B, Class overseas, unless otherwise specified by the contracting activity. Insofar as shipping container shall be of uniform shape, size, minimum tare and cube consistent with the protection required.

5.2.2. Level B. Horizontal situation indicator packaged as specified in 5.1.1 shall be packed in shipping containers conforming to PPP-B-636, class weather-resistant, unless otherwise specified by the contracting activity. Other requirements as specified in 5.2.1 apply.

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5.2.3 Level C. Packing shall be applied which affords protection from domestic shipment from the supply source to the first receiving activity for immediate use. This level shall conform to applicable carrier rules and regulations.

5.2.4. Industrial. The packaged horizontal situation indicator shall be packaged in accordance with ASTM D3951.

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5.3. Marking. In addition to any other markings required by the contract or order (see 6.2), interior and exterior containers shall be marked in accordance with MIL-STD-129."

* 6.2 Delete "title and first sentence" and add: "Acquisition requirements. Acquisition documents must specify the following:".

* Add: "6.2.e Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.10.)"

* 6.3 Delete entire paragraph and add: "6.3 First article. A first article is required for inspection and it shall be tested and approved under the appropriate provisions of 7-104.55 of the Defense Acquisition Regulation. First article tests are described in 4.5. The contracting officer should include specific instructions in all procurement instruments, regarding arrangements for examinations, test and approval of the first article.

* 6.5 Delete in its entirety.

The margins of this amendment are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous amendment were made. This was done as a convenience only and the government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous amendment.

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